

Case Resolution Assistant — Solution Design

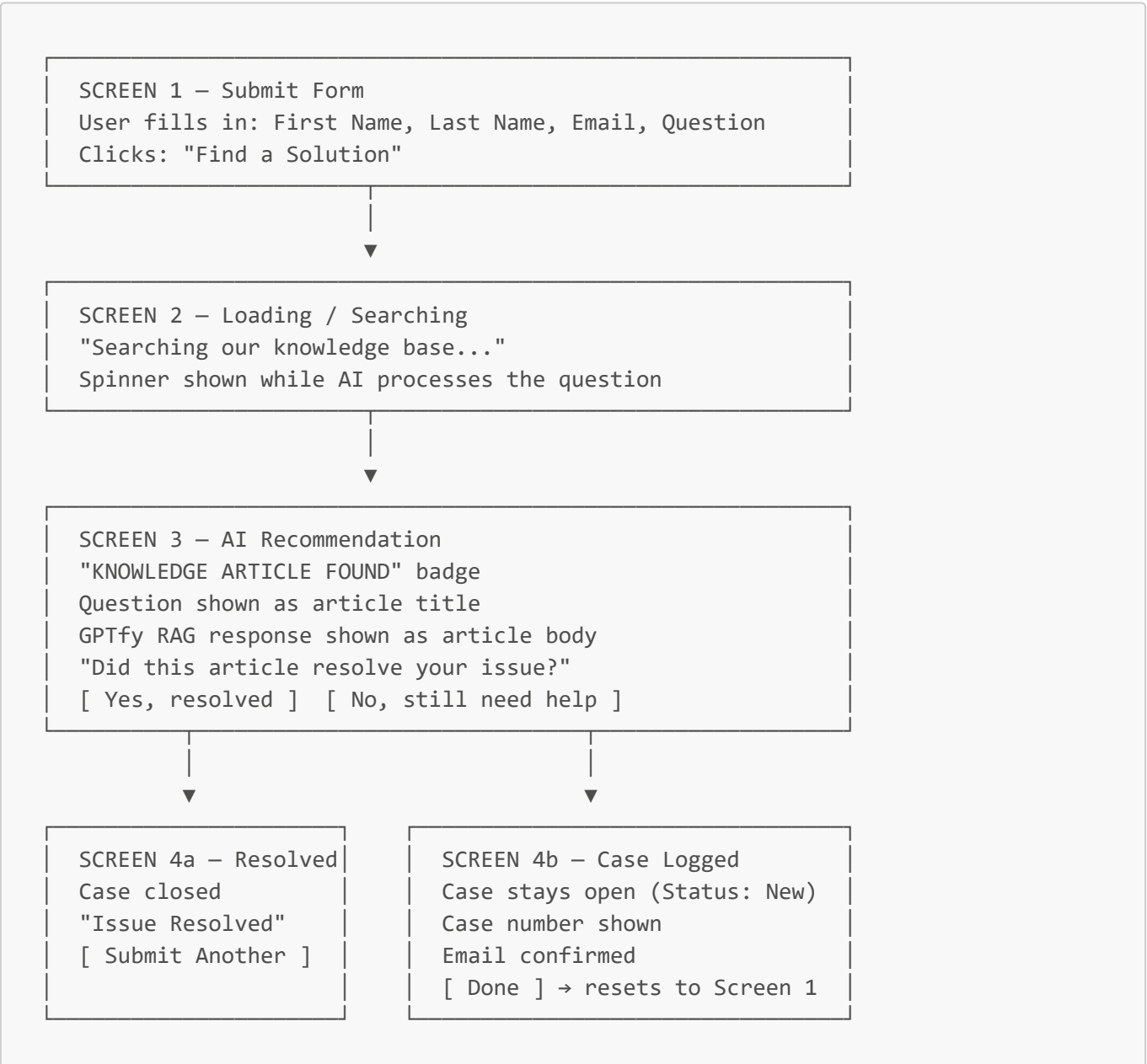
Overview

The **Case Resolution Assistant** is an AI-powered self-service support tool built on Salesforce Experience Cloud. When a user submits a question, the system creates a Salesforce Case and immediately triggers a background AI search using GPTfy's RAG (Retrieval-Augmented Generation) engine. The AI response is surfaced back to the user in real time. The user can then confirm whether the answer resolved their issue, which determines the final state of the case.

This solution is designed for a **public-facing Experience Cloud site** (no login required) and is suitable for demo and production use.

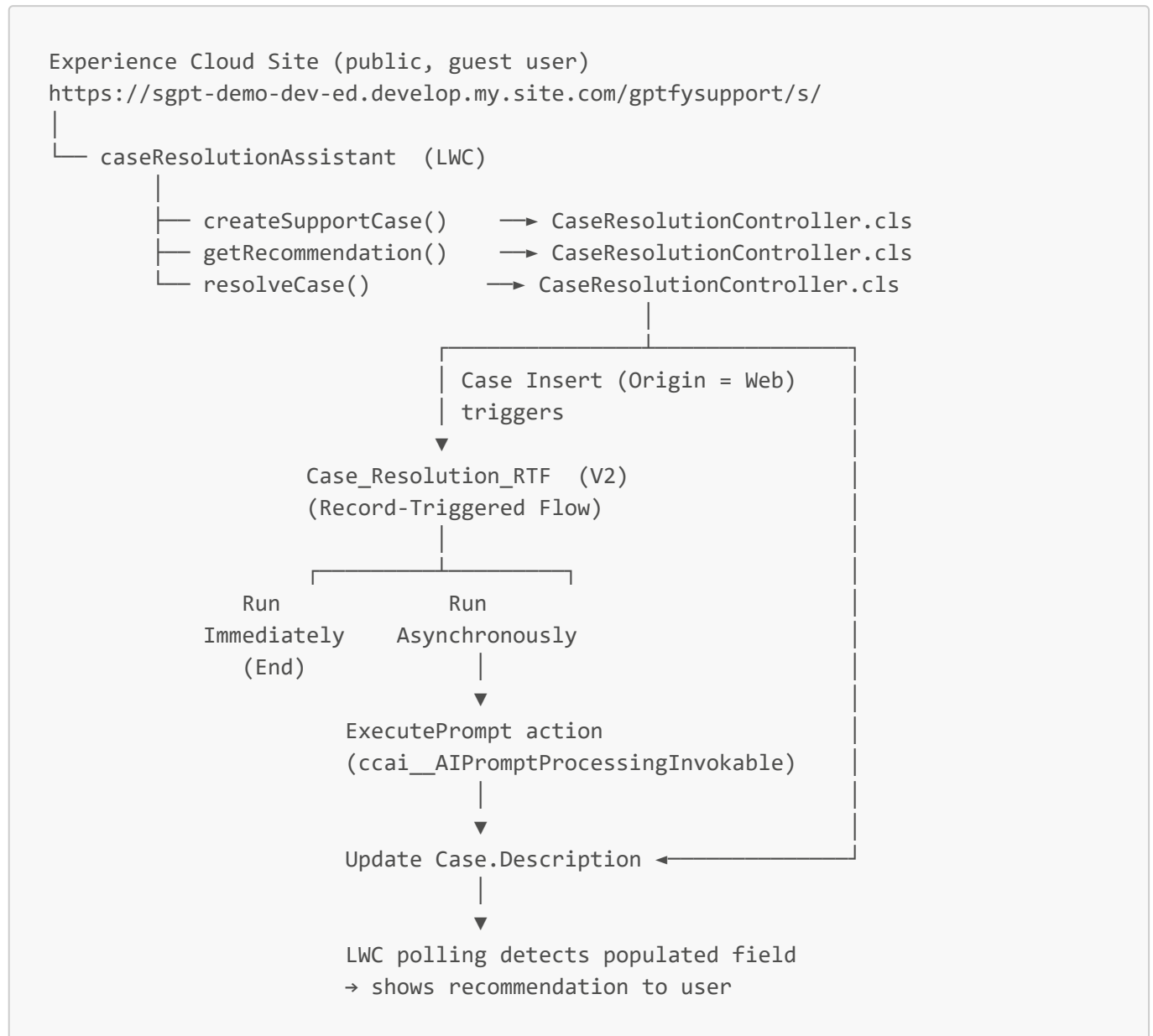
Live Site URL: <https://sgpt-demo-dev-ed.develop.my.site.com/gptfysupport/s/>

User Journey



Architecture

Component Map



Why Async?

Salesforce does not allow API callouts (HTTP requests to external systems) on a synchronous record-save transaction. The flow has two paths at the start:

- **Run Immediately** — exits immediately (no action needed in the sync transaction)
- **Run Asynchronously** — executes after the transaction commits, allowing the GPTfy external callout to proceed without restrictions

This keeps the Apex controller clean and avoids callout-in-transaction errors.

Components

1. LWC — caseResolutionAssistant

File	Purpose
caseResolutionAssistant.html	Template with 6 UI states, CSS-class-based visibility
caseResolutionAssistant.js	State machine, form handling, polling loop
caseResolutionAssistant.css	Full custom design matching GPTfy Support Community style
caseResolutionAssistant.js-meta.xml	Exposed to all Lightning targets including Experience Cloud

UI States

State constant	Visible when
INPUT	Page loads; after reset / Done
LOADING	Case creation complete, polling in progress
RECOMMENDATION	AI response received
RESOLVING	"Yes" clicked, closing case
RESOLVED	Case successfully closed
CASE_CREATED	"No" clicked, case stays open

Polling Logic

After case creation, the LWC calls `getRecommendation()` every **2.5 seconds** for up to **12 attempts (30 seconds total)**. Once `Case.Description` is non-blank, polling stops and the recommendation is displayed. If the timeout is reached with no response, a graceful fallback message is shown and the user is still presented with the Yes/No choice.

```
createSupportCase() → caseId
  ↳ poll every 2.5s:
      getRecommendation(caseId)
        → null? → keep polling
        → text? → show recommendation
        → 30s? → show fallback message
```

Visibility approach

All six state containers are always present in the DOM. Visibility is controlled by toggling `cra-state` vs `cra-state--hidden` CSS classes via computed getters. This ensures reliable rendering inside both the Experience Builder canvas and the published site.

2. Apex Controller — CaseResolutionController

without sharing — required to allow Experience Cloud guest users to perform DML on Case.

Method	Called by	Description
createSupportCase()	LWC on form submit	Inserts Case with Origin='Web', Status='New'. Returns caseId and caseNumber.
getRecommendation()	LWC polling loop	Reads Case.Description. Returns null if blank (still processing), returns text when ready.
resolveCase()	LWC on "Yes" click	Updates Case.Status = 'Closed'.

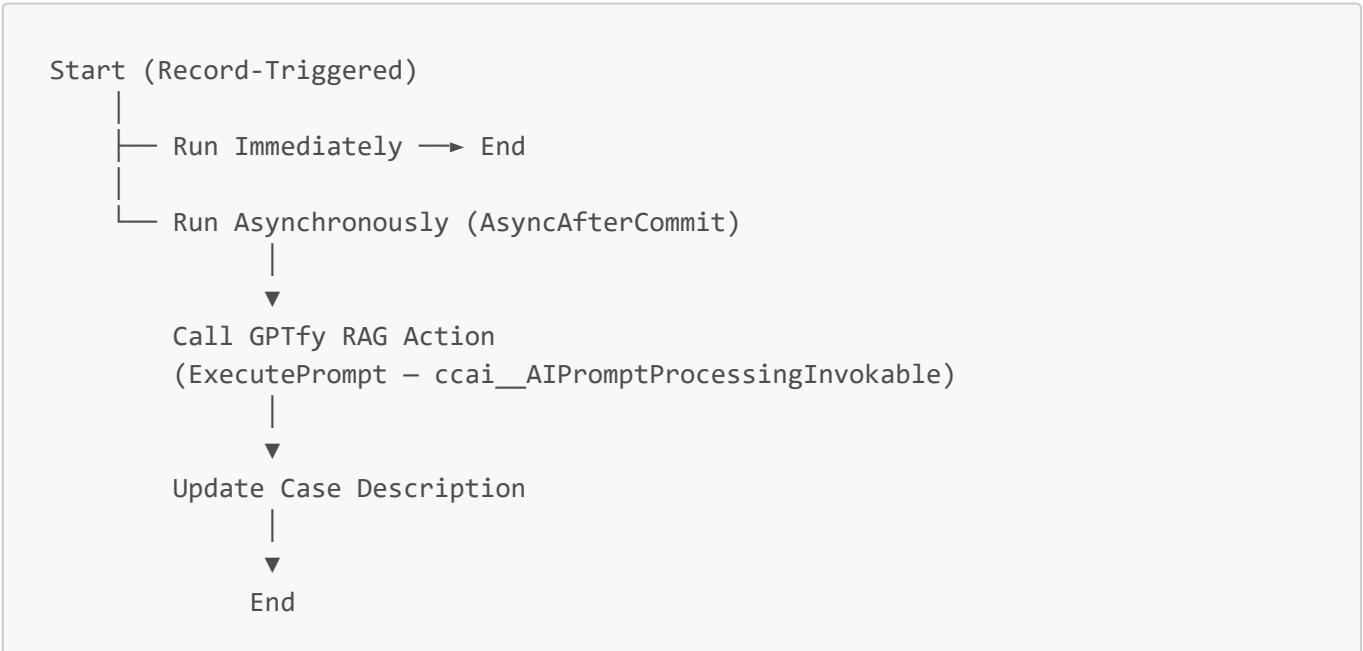
Case fields set on creation

Field	Value
Subject	User's question (max 255 chars)
SuppliedName	First Name + Last Name
SuppliedEmail	Email from form
Status	New
Origin	Web
Priority	Medium

3. Flow — Case_Resolution_RTF (V2)

Type: Record-Triggered Flow (After Save, Create only)
Trigger filter: Origin = Web
Active version: V2 (configured and debug-verified in Flow Builder)

Flow Structure



GPTfy Action Inputs (as configured in Flow Builder)

Input	Value	Source
EventUUID	Running Flow Interview → InterviewGuid	Flow system variable
promptRequestId	9cf7e95aacf31f5730a118c90c6fda4676843	Hardcoded in flow assignment
recordId	Triggering Case → Case ID	Record trigger context
customPromptCommand	Case Subject	Triggering Case → Subject

Note: EventUUID uses the Flow's built-in InterviewGuid (unique per flow execution) rather than the Case ID, which is the recommended approach for GPTfy event tracking.

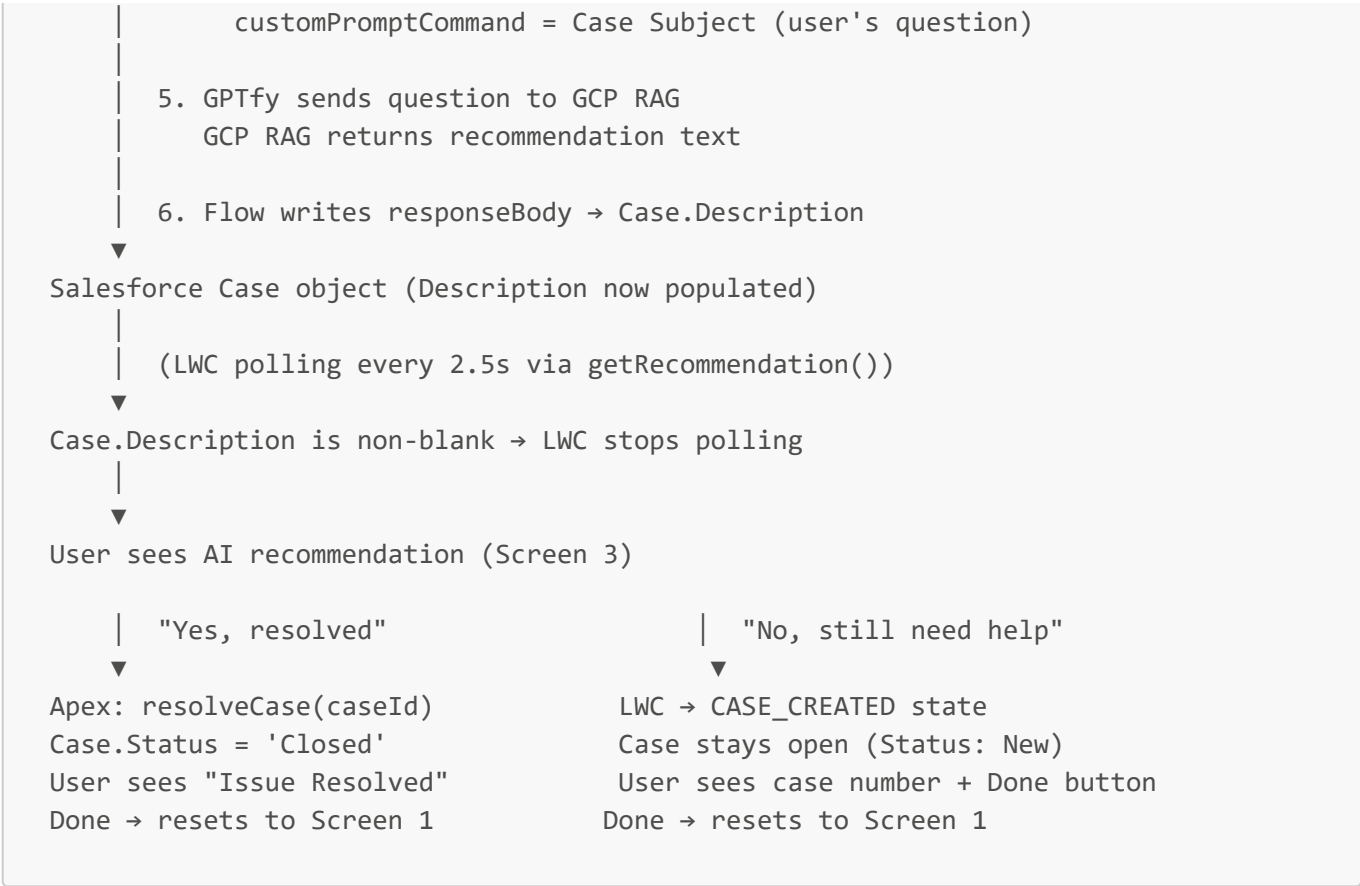
Flow Output

The responseBody returned by GPTfy is written to Case.Description. This is the field the LWC polls to detect when the AI response is ready.

Prompt configuration: The promptRequestId (9cf7e95aacf31f5730a118c90c6fda4676843) references the GPTfy Prompt record in this org. To change the prompt behaviour in future, update this value in the flow's assignment element — no code changes required.

Data Flow Diagram





Security Model

Layer	Configuration
Apex	without sharing — bypasses record-level sharing for guest users
Guest User Profile	Apex Class Access on CaseResolutionController
Guest User Profile	Create, Read, Edit on Case object
Case field access	Description field readable by guest profile (required for polling)
Site access	Experience Cloud site set to Public (no login required)

File Structure

```
force-app/main/default/
├── classes/
│   ├── CaseResolutionController.cls
│   └── CaseResolutionController.cls-meta.xml
├── flows/
│   └── Case_Resolution_RTF.flow-meta.xml
├── lwc/
│   └── caseResolutionAssistant/
│       ├── caseResolutionAssistant.html
│       └── caseResolutionAssistant.js
```

└─ caseResolutionAssistant.css

└─ caseResolutionAssistant.js-meta.xml

Polling Timeout Behaviour

If GPTfy does not respond within 30 seconds (12 polls × 2.5s), the LWC stops polling and displays:

"Our AI is still processing your request. A support agent will review your case and reach out to you shortly."

The Case is already created at this point. The user sees the Yes/No choice with the fallback text. If they click "No", the case number is shown with the Done button.

Design Decisions

Decision	Rationale
Case created before AI call	GPTfy's <code>ccai__AIPromptProcessingInvokable</code> requires a <code>recordId</code> . Creating the case first provides this and ensures a case always exists regardless of AI outcome.
Async Flow path (Run Asynchronously)	Salesforce prohibits HTTP callouts in synchronous DML transactions. The async path avoids this constraint. The "Run Immediately" path exits with no action.
InterviewGuid for EventUUID	The Flow's built-in <code>InterviewGuid</code> is unique per execution and is the recommended identifier for GPTfy event tracking, replacing the earlier Case ID approach.
Polling over Platform Events	Simpler to implement for guest users on Experience Cloud. Platform Events require CometD subscriptions which are complex for unauthenticated users.
Case.Description for AI response	Avoids creating a custom field. Description is a standard long-text field suitable for storing the AI recommendation and readable in the polling query.
CSS-class visibility over <code>lwc:if</code>	Experience Builder canvas renders all <code>lwc:if</code> content simultaneously. CSS <code>display:none !important</code> ensures only the active state is visible in both canvas and published modes.
Prompt ID hardcoded in Flow assignment	Keeps prompt configuration change in Flow Builder only — no code deployment needed to update the GPTfy prompt in future.